

Fig. 3: An actual-size PCB layout for the transmitter

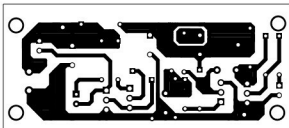


Fig. 5: An actual-size PCB layout for the receiver

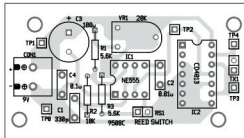


Fig. 4: Component layout for transmitter PCB

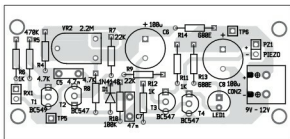


Fig. 6: Component layout for receiver PCB

and buzzer PZ1. Power supply for the circuit can be from 9V to 12V.

Construction and testing

An actual-size, single-side PCB for the

transmitter is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an actual-size, single-side PCB layout for the receiver is shown in Fig. 5 and its component layout in Fig. 6.

After assembling the transmitter and receiver units on two separate PCBs, enclose them in separate plastic cases.

When the door is opened, TX1 emits 40kHz signal, which RX1 receives and activates the buzzer wirelessly. Transmitter TX1 must be directed towards receiver transducer RX1 for proper functioning. Adjust VR1 in the transmitter to obtain maximum range. The range was found to be

around 1 metre. In case of any problem, verify voltages at the test points given in the table. ●

The author is a regular contributor to EYF

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